

TERNOPIL, Ukraine

WMO: 334150

Lat: 49.5278N

Lon: 25.6911E

Elev: 1079

StdP: 14.13

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.0	4.5	-6.3	4.1	-1.5	-0.2	5.7	5.1	26.6	32.4	23.0	31.1	4.0	20	0.595

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	85.3	67.7	82.2	66.3	79.3	65.1	69.8	81.3	68.2	78.7	66.6	76.3	9.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	99.2	74.7	64.3	94.0	73.0	62.9	89.4	71.4	34.4	81.6	33.1	78.7	31.8	76.6	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.9	19.9	18.3	-7.5	89.8	7.7	2.6	-13.1	91.7	-17.6	93.2	-22.0	94.7	-27.6	96.6		
			-7.9	72.4	7.7	1.5	-13.4	73.4	-17.9	74.3	-22.2	75.1	-27.7	76.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.5	24.7	27.4	35.5	48.0	57.5	63.0	66.5	65.5	56.7	46.6	37.0	27.8
	DBStd	16.84	10.23	10.49	8.77	7.62	6.67	5.74	5.31	5.48	6.49	7.53	8.20	9.51
	HDD50	3258	784	632	451	122	14	0	0	0	15	157	393	690
	HDD65	7005	1249	1052	914	509	245	105	47	60	258	572	839	1155
	CDD50	1966	0	0	2	63	247	391	511	481	216	50	5	0
	CDD65	239	0	0	0	1	12	46	93	77	10	0	0	0
	CDH74	2238	0	0	0	15	151	373	820	752	126	1	0	0
	CDH80	519	0	0	0	2	15	51	213	217	21	0	0	0
Wind	WSAvg	8.8	9.5	9.8	10.0	9.4	8.8	8.3	7.7	7.2	8.1	8.6	9.4	9.3
Precipitation	PrecAvg	24.0	1.3	1.3	1.1	1.7	2.7	3.1	3.3	2.5	2.2	1.3	1.5	1.5
	PrecMax	32.1	4.2	3.2	2.7	4.5	4.8	7.2	8.2	7.1	7.1	3.9	5.8	2.6
	PrecMin	15.6	0.2	0.1	0.2	0.3	0.7	0.7	0.4	0.6	0.1	0.2	0.0	0.2
	PrecStd	4.0	1.0	0.8	0.7	1.1	1.1	1.5	1.6	1.2	1.6	0.9	1.0	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.3	52.3	61.9	75.8	82.3	84.5	89.3	89.7	82.8	72.4	60.4	48.1
		MCWB	41.8	45.2	49.8	58.4	64.0	67.8	69.9	68.1	64.2	59.3	52.7	44.3
	2%	DB	41.8	45.9	55.6	69.8	78.3	81.3	85.6	85.6	77.5	66.7	54.7	44.6
		MCWB	38.5	41.9	46.7	55.6	62.2	66.2	68.4	67.2	61.6	56.5	50.4	42.3
	5%	DB	39.0	42.3	51.0	65.4	74.7	78.5	82.0	81.9	73.2	61.7	51.2	41.1
		MCWB	36.6	39.3	43.8	53.3	60.5	65.0	67.0	66.3	59.9	54.3	48.3	39.2
	10%	DB	36.4	39.1	47.2	61.2	70.8	75.4	78.7	78.0	68.9	57.7	48.3	38.0
		MCWB	34.5	36.6	42.2	50.7	58.9	64.1	65.8	64.9	58.6	52.4	45.9	36.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.5	46.1	50.6	59.5	66.4	70.6	72.3	71.8	65.9	61.2	53.4	45.6
		MCDB	45.5	52.1	59.9	72.9	78.1	80.8	84.1	83.7	79.6	69.7	58.4	47.6
	2%	WB	39.0	42.1	47.4	56.8	63.8	68.3	70.1	69.4	63.3	57.5	51.0	42.2
		MCDB	41.2	45.5	54.4	68.0	74.7	78.0	81.8	81.5	73.2	64.5	54.3	44.2
	5%	WB	36.8	39.4	44.8	54.2	61.9	66.6	68.4	67.5	61.4	55.2	48.6	39.2
		MCDB	38.7	42.1	49.9	63.1	72.2	75.6	79.1	78.2	70.1	60.6	51.1	41.0
	10%	WB	34.7	36.8	42.4	52.0	59.8	64.9	66.9	65.8	59.4	52.7	45.9	36.6
		MCDB	36.2	38.9	46.7	59.8	68.8	73.1	76.4	75.3	67.5	57.2	48.0	37.9
Mean Daily Temperature Range	5% DB	MDBR	8.3	9.7	12.3	16.9	18.1	17.5	18.1	19.2	17.0	14.2	9.2	8.2
		MCDBR	8.6	13.0	20.2	24.2	24.7	23.2	24.4	26.2	25.4	21.9	15.3	10.6
		MCWBR	6.8	9.5	12.6	12.5	11.4	10.6	10.0	10.7	11.7	12.6	10.7	8.6
	5% WB	MCDBR	7.9	12.6	17.9	21.7	21.9	19.8	21.4	22.7	21.9	19.3	14.1	10.0
		MCWBR	6.6	9.6	11.8	11.9	10.7	9.9	9.6	10.3	11.1	11.7	10.3	8.3
Clear-Sky Solar Irradiance	taub	0.309	0.339	0.379	0.430	0.412	0.411	0.439	0.439	0.399	0.364	0.340	0.306	
	taud	2.238	2.195	2.192	2.126	2.247	2.269	2.201	2.199	2.269	2.350	2.339	2.309	
	Ebn at Noon	228	249	259	258	269	270	259	252	250	239	216	213	
	Edh at Noon	23	32	38	45	42	41	43	42	35	27	22	19	
All-Sky Solar Radiation	RadAvg	294	554	913	1338	1686	1796	1734	1550	1077	666	301	208	
	RadStd	46	79	120	139	166	130	129	127	166	99	35	35	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.17	N/A	N/A	+1.53	+1.01	N/A	N/A	-374	+216	+83
(47)	Regional (10 neighbors)	+1.17	N/A	N/A	+1.71	+0.83	N/A	N/A	-360	+212	+86

Nomenclature: See separate page